

AGGRCOW - Aggressive cows

#binary-search

Farmer John has built a new long barn, with N ($2 \leq N \leq 100,000$) stalls. The stalls are located along a straight line at positions $x_1 \dots x_N$ ($0 \leq x_i \leq 1,000,000,000$).

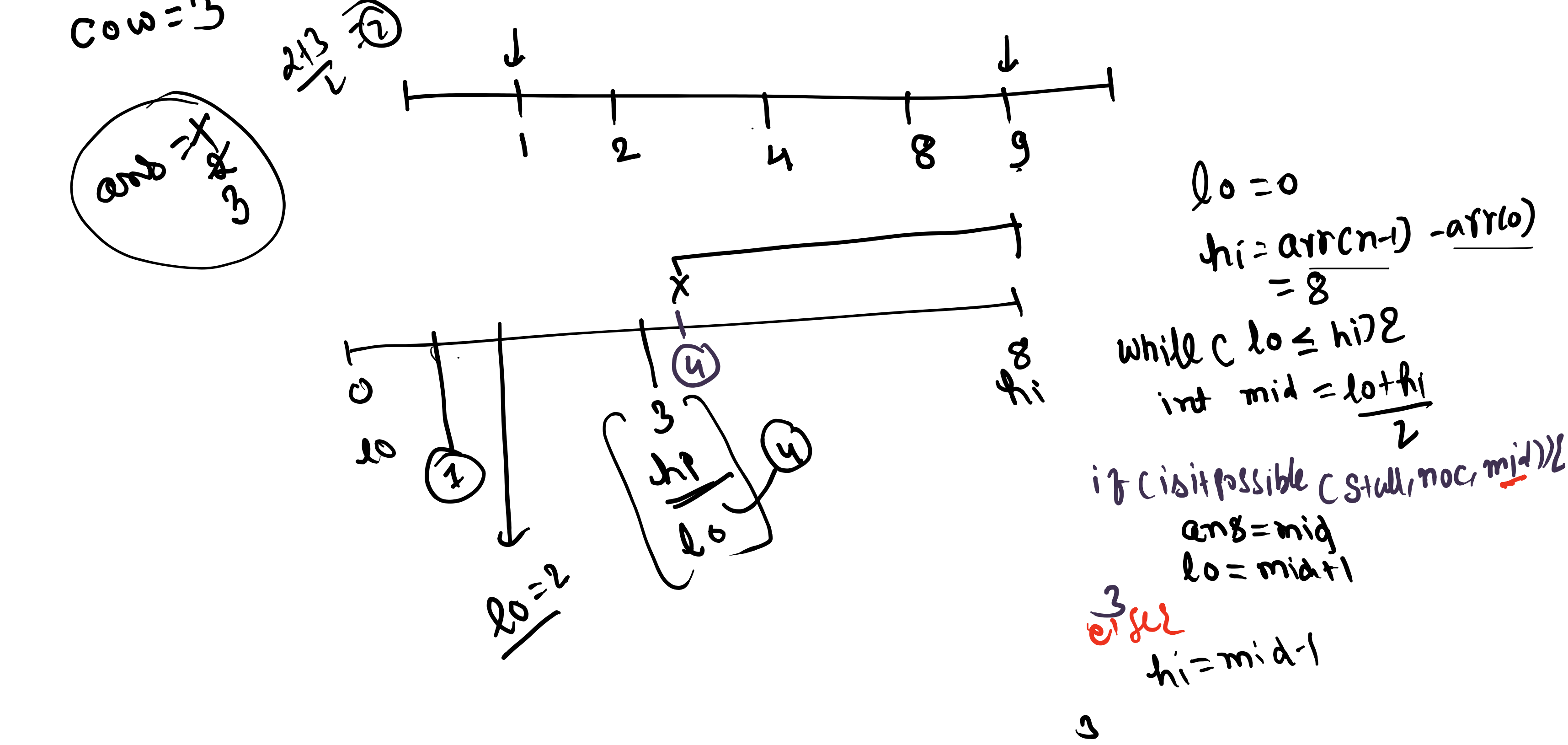
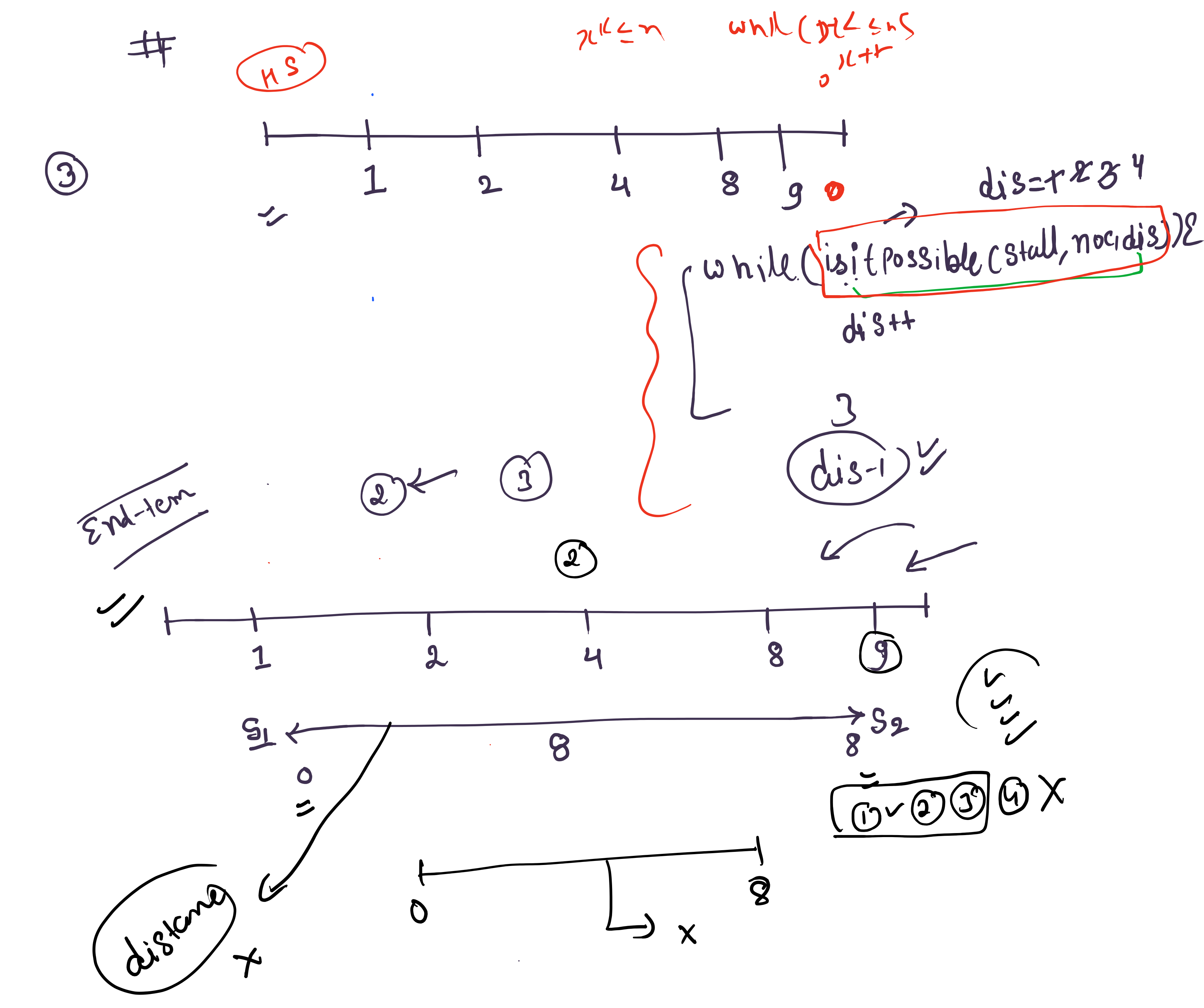
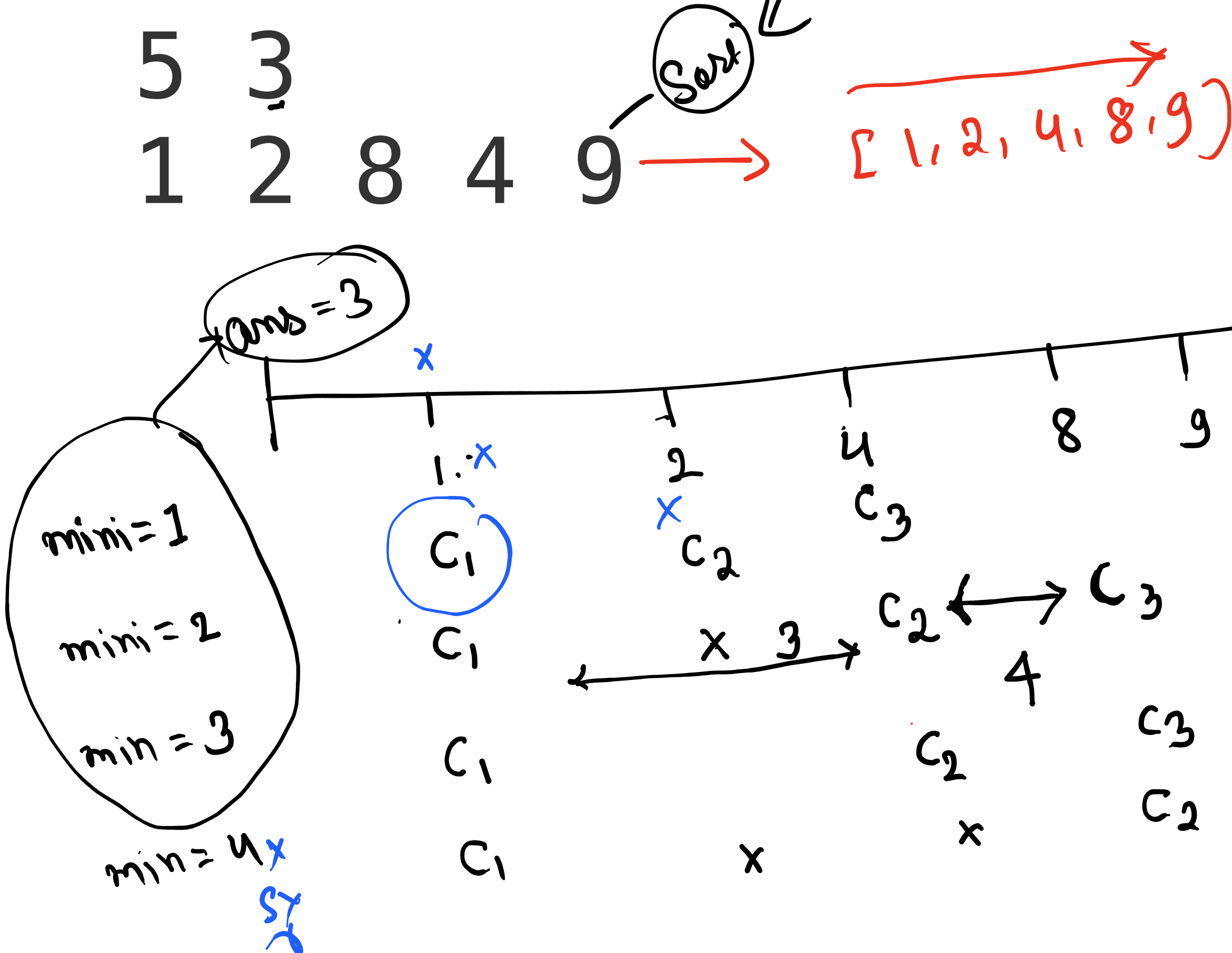
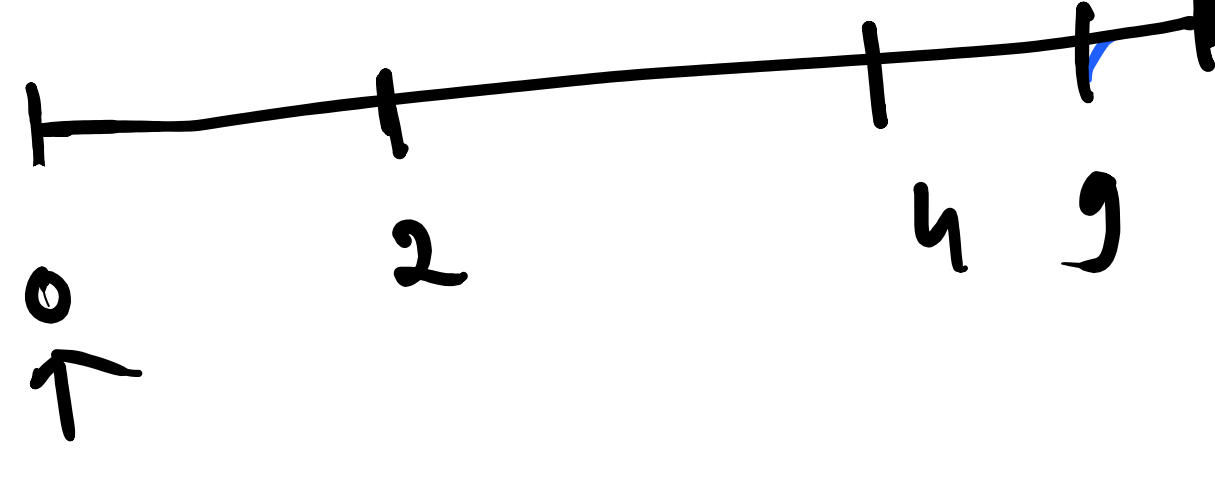
His C ($2 \leq C \leq N$) cows don't like this barn layout and become aggressive towards each other once put into a stall. To prevent the cows from hurting each other, FJ wants to assign the cows to the stalls, such that the minimum distance between any two of them is as large as possible. What is the largest minimum distance?

[4,2,9]

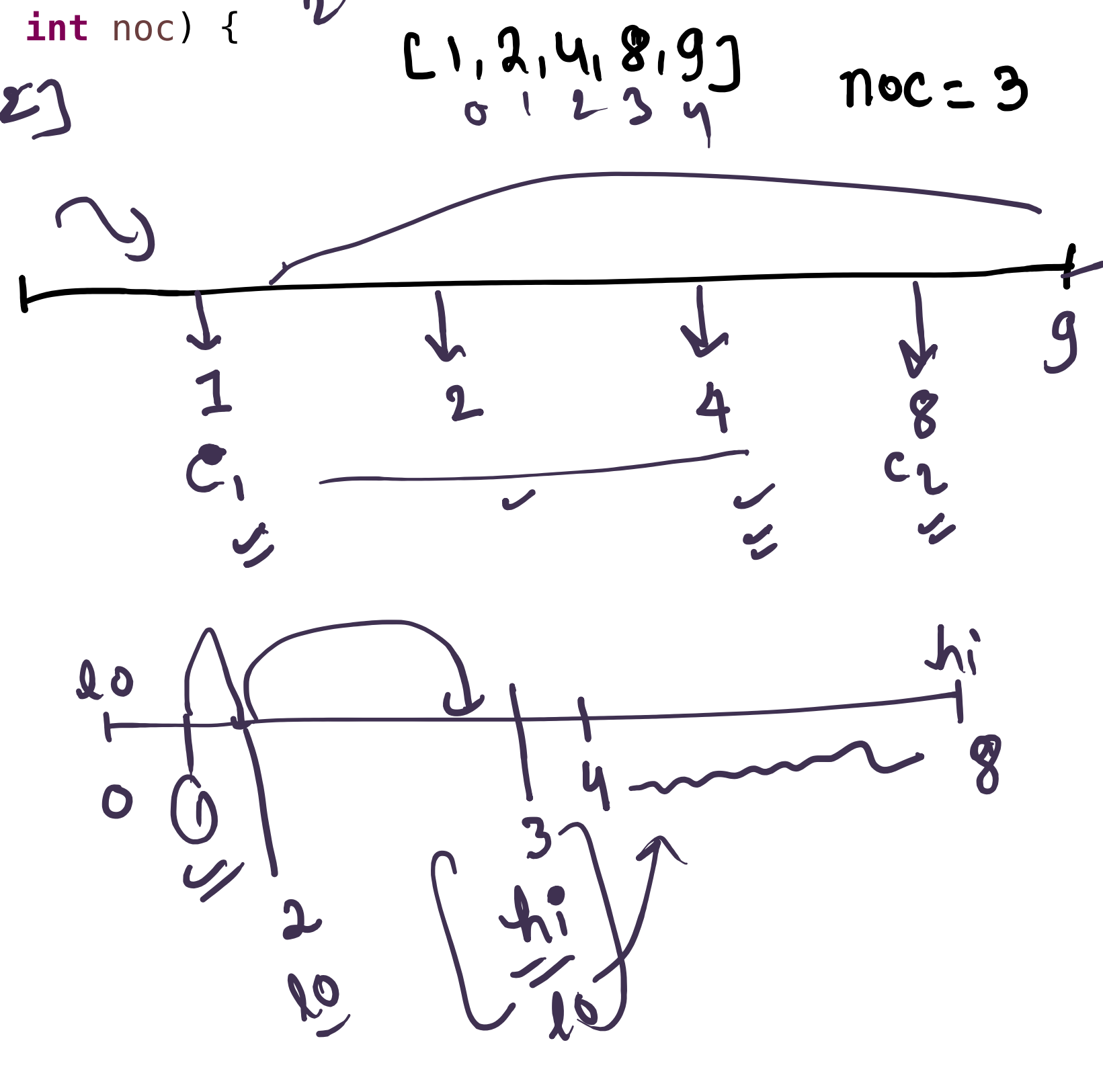
Input

t - the number of test cases, then t test cases follows.

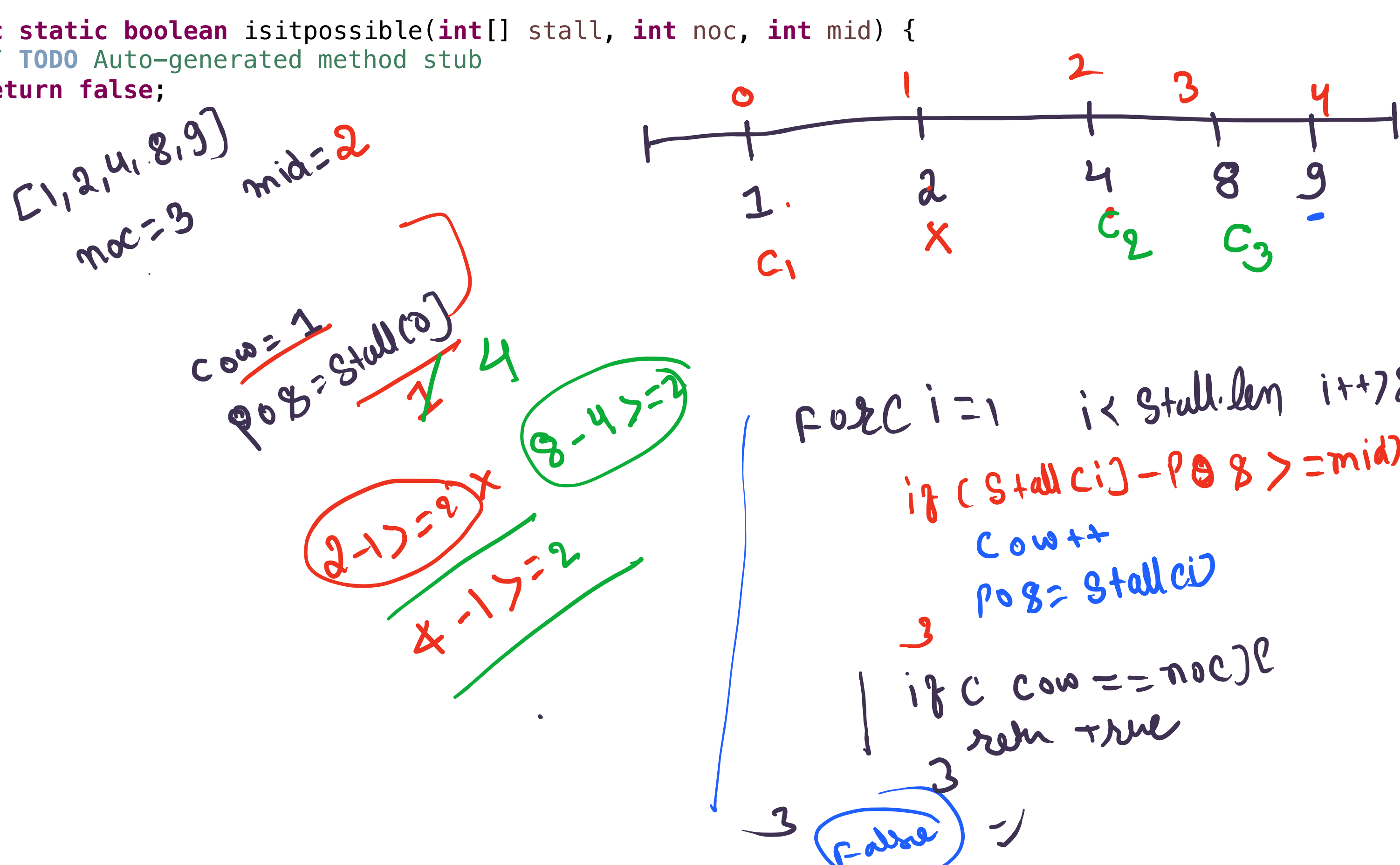
- Line 1: Two space-separated integers: N and C
- Lines 2..N+1: Line i+1 contains an integer stall location, x_i



```
public static int largest_minimum_distance(int[] stall, int noc) {
    int ans = 0, lo = 0;
    int n = stall.length;
    int hi = stall[n - 1] - stall[0];
    while (lo <= hi) {
        int mid = (lo + hi) / 2;
        if (ispossible(stall, noc, mid)) {
            ans = mid;
            lo = mid + 1;
        } else {
            hi = mid - 1;
        }
    }
    return ans;
}
```



```
public static boolean isitpossible(int[] stall, int noc, int mid) {
    // TODO Auto-generated method stub
    return false;
}
```



Book Allocation Problem

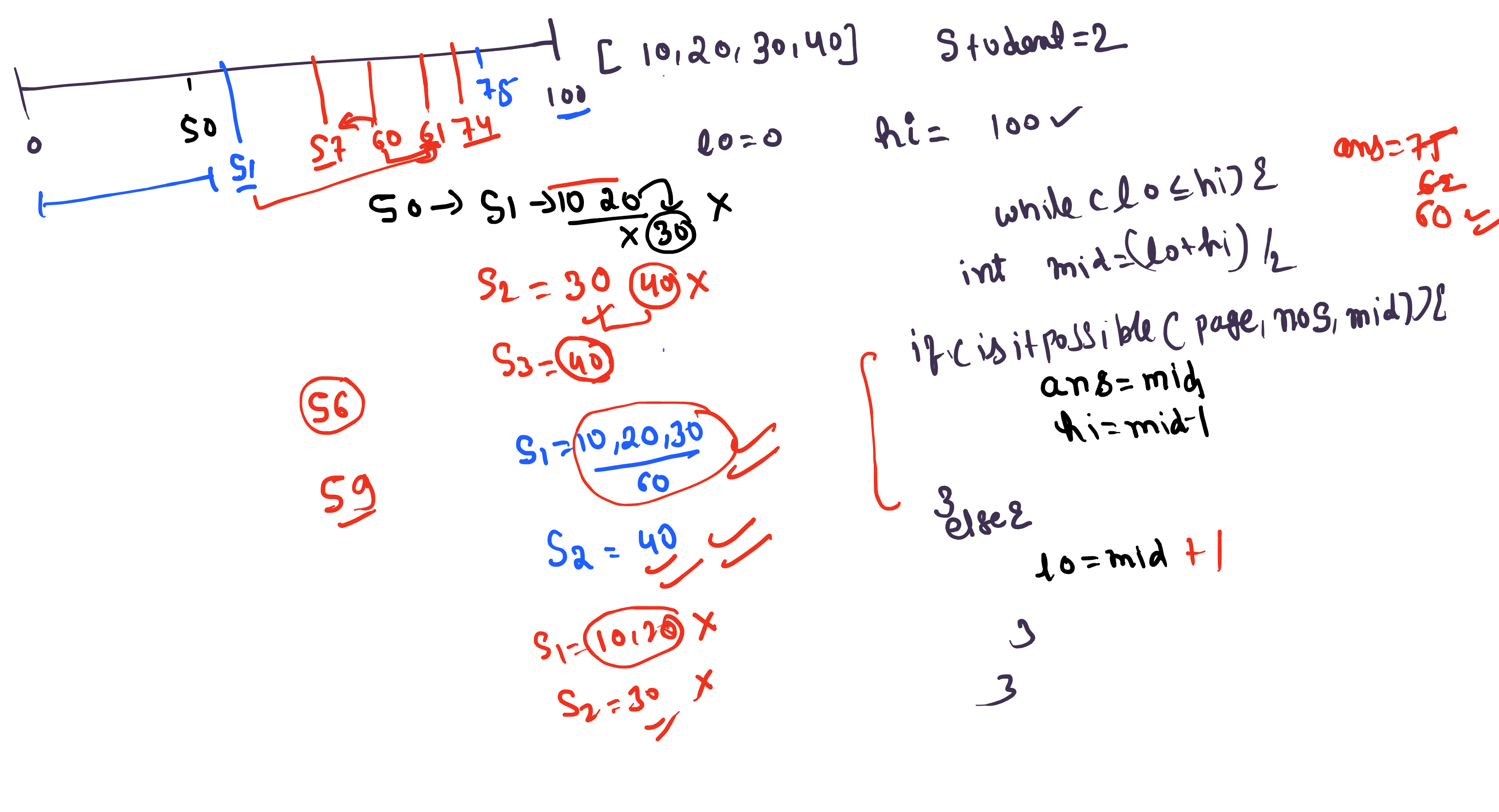
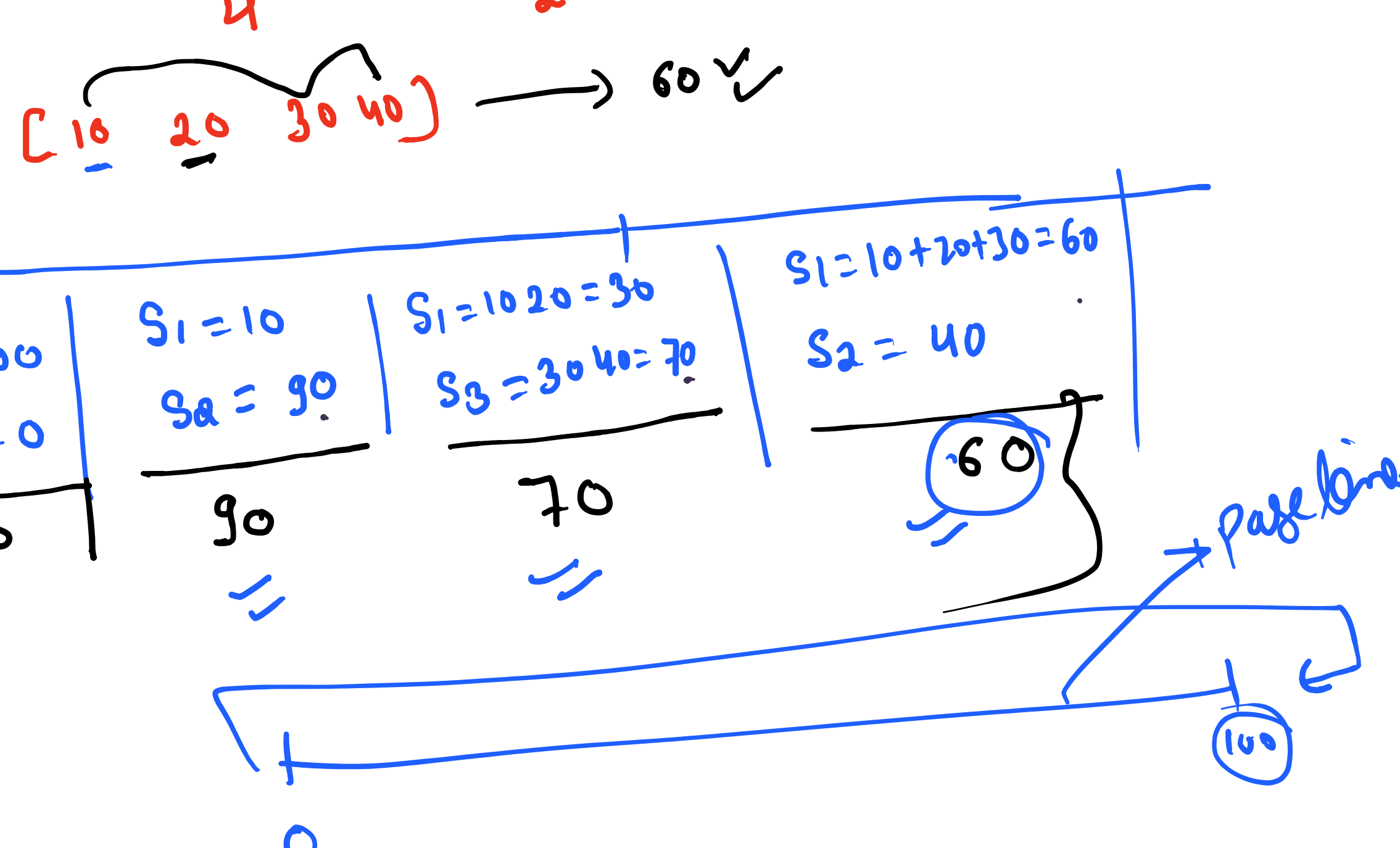
You are given number of pages in n different books and m students. The books are arranged in ascending order of number of pages. Every student is assigned to read some consecutive books. The task is to assign books in such a way that the maximum number of pages assigned to a student is minimum.

Input format

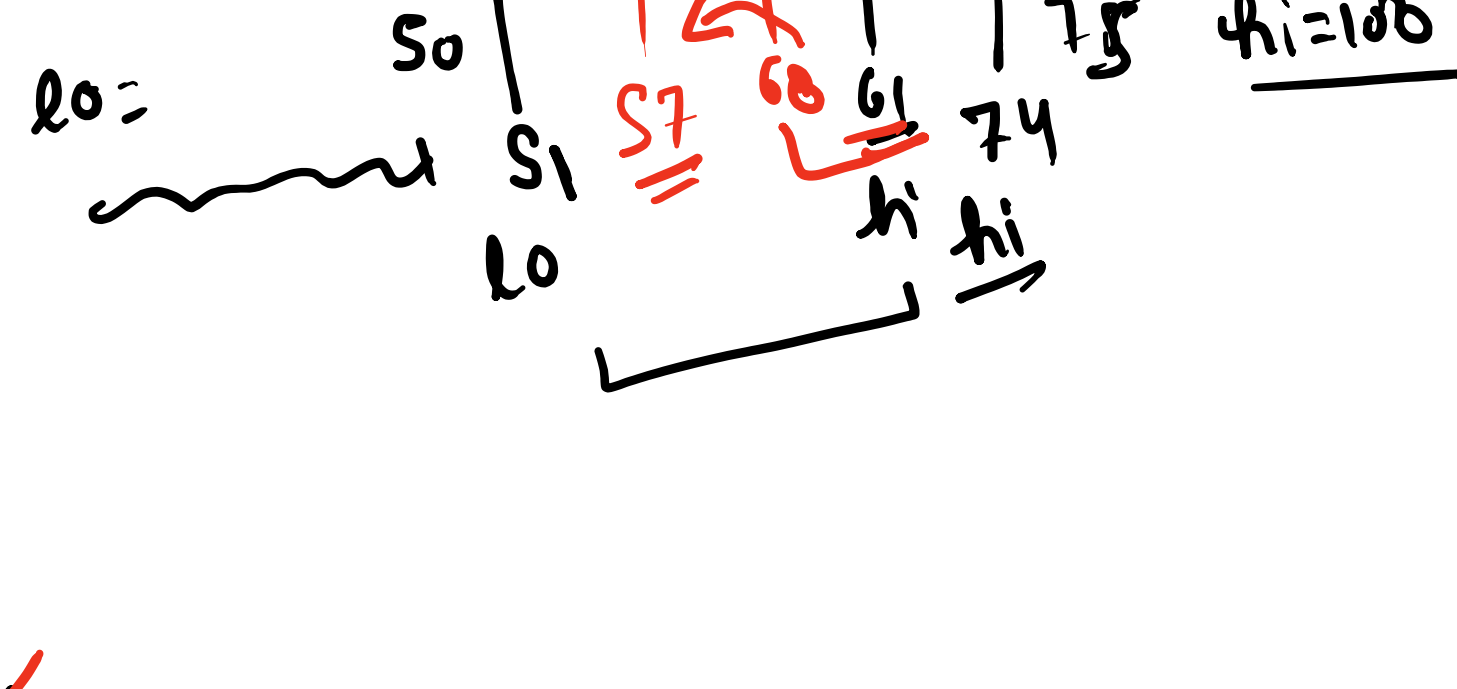
First line contains integer t as number of test cases.
Next t lines contains two lines.
For each test case, 1st line contains two integers n and m which represents the number of books and students and 2nd line contains n space separated integers which represents the number of pages of n books in ascending order.

Output format

Print the maximum number of pages that can be assigned to students.



```
public static int minimum_Page(int[] page, int nos) {
    int lo = 0, hi = 0;
    for (int i = 0; i < page.length; i++) {
        hi = hi + page[i];
    }
    int ans = 0;
    while (lo <= hi) {
        int mid = (lo + hi) / 2;
        if (ispossible(page, nos, mid)) {
            ans = mid;
            hi = mid - 1;
        } else {
            lo = mid + 1;
        }
    }
    return ans;
}
```



```
public static boolean isitpossible(int[] page, int nos, int mid) {
    // TODO Auto-generated method stub
    return false;
}
```

